

Statistical Methods for Quantum State Tomography: Evaluating Maximum Likelihood Estimation and Variational Autoencoders

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Abstract—Quantum State Tomography (QST) is a crucial technique in quantum computing. State reconstruction helps diagnose problems in a circuit and supports efforts to improve it. However, traditional approaches suffer from severe scalability problems and for this reason generative models can be considered as a way to address these issues. This work aims to investigate the limitations of classical QST techniques and to assess whether generative models — specifically a Variational Autoencoder (VAE) — can offer a more scalable alternative to a standard Maximum Likelihood Estimator (MLE).

We analyze three types of states — the GHZ state, the W state, and a product state — and we compare the classical fidelity of MLE and VAE reconstructions for 3-qubit system while scaling the number of measurements. We also show that the computational cost of MLE becomes intractable for higher qubit counts, and present VAE results up to an 8-qubit state on the GHZ state.

In the 3-qubit comparison, MLE outperformed VAE for the entangled states (GHZ and W), while both methods had similar results on the product state. However, already at 4 qubits MLE required several minutes of computation, while VAE took only a few seconds. At 8 qubits, VAE maintained strong generalization capabilities, achieving high classical fidelity with a limited number of training samples.

We conclude that MLE is an effective technique for QST at low qubit numbers; however, its poor scalability shows that VAE is an overall more practical method, achieving high performance with a limited number of measurements on the state.

1. INTRODUCTION

Nowadays, reconstruction of quantum state is fundamental in quantum computing to check how well a state was prepared in the laboratory and where the circuit needs improvement. In the Noisy Intermediate Scale Quantum (NISQ) era, where noise is a crucial property of quantum hardware, this verification is critical; researchers rely on it to certify entangled state preparation, benchmark logic gate fidelity, and validate noise mitigation strategies before scaling up circuit depth or qubit number. The standard technique for this task is Quantum State Tomography (QST), which consists in the reconstruction of a quantum state from a set of measurement outcomes.

QST, however, suffers from an important dimensionality problem: the Hilbert space, where quantum states live, scales exponentially with the number of qubits, meaning that a complete characterization of an N -qubit system requires estimating $4^N - 1$ independent parameters. As a direct consequence, both the number of measurements needed and the computational cost of classical estimators, such as Maximum Likelihood Estimation (MLE), grow exponentially with the qubits count, making classical reconstruction impractical just beyond a bunch of qubits. In this context, data-driven generative models offer a promising alternative: rather than reconstructing the full density matrix, they learn the underlying structure of the measurement outcome distribution directly from data, bypassing the exponential parameter scaling entirely.

Among generative approaches, Chen et al. [5] used a Variational Autoencoder (VAE) trained on IC-POVM measurement outcomes to reconstruct GHZ states up to 8 qubits, showing that the number of training samples required to reach 0.99 fidelity scales approximately linearly with qubit number for GHZ states. This work is built directly on the approach just described and extended it with a systematic comparison against MLE and testing performance across three representative state families: GHZ, W, and product states. We apply both methods to simulated IC-POVM measurement data at low number of qubits, comparing classical fidelity and computational cost.

This report is structured as follows. [Section 2](#) introduces the theoretical background, starting from the basics of quantum information theory and quantum measurement theory, then covers the Maximum Likelihood Estimation, and finally discusses the role of generative modeling and VAEs. [Section 3](#) explains how the theory was implemented in this work and describes the structure of the VAE. [Section 4](#) presents the results for MLE and VAE, along with the comparative analysis conducted.

2. THEORETICAL BACKGROUND

Here a brief theoretical introduction to the most important concepts used will be given. More specifically, basics of quantum measurement theory will be introduced, with particular focus on POVMs; we will then explain the statistical methods behind the MLE technique; and finally, generative deep learning methods will be discussed, with particular attention to VAEs.

2.1. Quantum States and POVM Measurements

In quantum computing, the fundamental unit of information is a two-level quantum mechanical system called *qubit* [1]. Unlike a classical bit, a qubit can exist in a state of $|0\rangle$, $|1\rangle$, or any linear superposition of these computational basis states:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (1)$$

where $\alpha, \beta \in \mathbb{C}$ provide the probability amplitude of these two parts and then $|\alpha|^2 + |\beta|^2 = 1$. Such states are called *pure states* and in case of single qubit system can be visually mapped onto the surface of the Bloch sphere as points with a unit radius ($r = 1$). However, these state vectors are only sufficient to describe closed, isolated quantum systems. In practical applications, due to the fragile nature of quantum states and their inevitable interactions with the surrounding environment, real-world devices suffer from noise and decoherence. This open system dynamics causes pure states to lose their quantum coherence, pulling the state vector inside the Bloch sphere ($r < 1$) and transforming it into a *mixed state*.

For this reason — in quantum information theory — the state of a physical system is very often described using an ensemble of states expressed by its density operator [1, 8], or density matrix

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i| \quad (2)$$

where p_i represents the classical probability of the system being prepared in the state $|\psi_i\rangle$. Under this formalism, a *pure state* is simply defined as $\rho = |\psi\rangle \langle \psi|$, while a *mixed state* occurs when no single pure state can describe the ensemble. To be physically valid, a density matrix must satisfy three constraints:

$$\rho^\dagger = \rho \quad (\text{Hermiticity}), \quad \text{Tr}(\rho) = 1 \quad (\text{unit trace}), \quad \rho \geq 0 \quad (\text{positive semi-definiteness}).$$

In this context, reconstructing a quantum state consists in the estimation of $d^2 - 1$ free real parameters of the relative density matrix, where d is the dimension of the Hilbert space and $d = 2^N$ for an N -qubit system.

The only way to extract information from these states is to perform multiple measurements. Traditionally, Quantum State Tomography relies on the standard Projection-Valued Measures (PVMs) and consists in measuring multiple distinct bases — such as the Pauli X , Y , and Z directions — to capture all the system's degrees of freedom. However, this approach scales poorly and requires multiple experimental runs to evaluate a single quantum state.

To overcome this limitation, a generalized measurement framework is mathematically described by a Positive Operator-Valued Measure (POVM). A POVM is defined as a set of positive semi-definite operators $\{M_a\}$ acting on the state space, where each index a corresponds to a specific experimental outcome. These operators must satisfy the completeness relation:

$$\sum_a M_a = \mathbb{I} \quad (3)$$

where $\mathbb{I}$ is the identity operator. For a single qubit, we employ a Symmetric Informationally Complete POVM (SIC-POVM), which consists of 4 projectors $\{M_\alpha\}_{\alpha=1}^4$ directed towards the vertices of a regular tetrahedron inside the Bloch sphere (see [Appendix A](#)). SIC-POVMs are useful for QST because they yield exactly 4 possible outcomes and then provides the minimal necessary information¹ to reconstruct a state without statistical redundancy [5].

For an N -qubit system, the global POVM can be constructed by taking the tensor product of local SIC-POVMs chosen for each individual qubit. In this context, if we index a global outcome by a vector $\vec{a} = (a_1, a_2, \dots, a_N)$, where each $a_i \in \{0, 1, 2, 3\}$ denotes the local outcome obtained on the i -th qubit, the corresponding global POVM operator $M_{\vec{a}}$ is defined as:

$$M_{\vec{a}} = M_{a_1} \otimes M_{a_2} \otimes \dots \otimes M_{a_N} = \bigotimes_{i=1}^N M_{a_i} \quad (4)$$

Finally, when the POVM $\{M_{\vec{a}}\}$ is performed on a state ρ , the probability of observing the outcome $\vec{a}$ is given by

$$p_{\vec{a}} = p(\vec{a}|\rho) = \text{Tr}(M_{\vec{a}}\rho) \quad (5)$$

which represents the generalized Born rule for POVM measurements. In this sense, by utilizing an IC POVM framework, complete information about the quantum state can be extracted within a single, unified measurement setup, avoiding the need for multiple basis reconfigurations [5, 8].

Once we have reconstructed a quantum state, we need quantitative metrics to evaluate how accurately it matches the true target state. In this work, we employ two complementary fidelity measures.

Quantum Fidelity. For two quantum states described by density matrices ρ and σ , the quantum fidelity is defined as [1, 8]:

$$F_Q(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2 \quad (6)$$

¹For $N = 1$, we need $2^2 - 1 = 3$ free parameters: 4 (outcome probabilities) - 1 (normalization constraint).

F_Q ranges from 0 to 1 and provides a direct measure of overlap in Hilbert space. However, it requires explicit knowledge of the density matrix ρ , making it applicable only to MLE, where ρ is explicitly reconstructed.

Classical Fidelity. Since the VAE operates directly on measurement outcome distributions without reconstructing ρ , we adopt the Bhattacharyya coefficient as a classical fidelity measure [5, 8]:

$$F_C(p, q) = \sum_{\vec{a}} \sqrt{p_{\vec{a}} q_{\vec{a}}} \quad (7)$$

where $p_{\vec{a}}$ and $q_{\vec{a}}$ are the probability distributions over POVM outcomes of the reconstructed and target states, respectively. F_C also ranges from 0 to 1 and serves as the primary metric throughout this work, as it enables a fair and direct comparison between MLE and VAE without requiring the full density matrix reconstruction.

2.2. Statistical Foundations of MLE

The Born Rule (Equation 5) consists of a linear map from the density matrix ρ to a probability vector P containing the probabilities of all possible global outcomes $\vec{a}$. In principle, in the limit of a large number of measurement samples ($N_s \rightarrow \infty$), one can invert this map to reconstruct the quantum state by substituting the theoretical probabilities with the empirical outcome frequencies $f_{\vec{a}} \approx p_{\vec{a}}$. Mathematically, this approach yields [5]:

$$\rho = \sum_{\vec{a}, \vec{a}'} f_{\vec{a}} (O^{-1})_{\vec{a}, \vec{a}'} M_{\vec{a}'} \quad \text{where} \quad O_{\vec{a}, \vec{a}'} = \text{Tr}(M_{\vec{a}} M_{\vec{a}'}) \quad (8)$$

where $\vec{a}, \vec{a}' = (a_1, a_2, \dots, a_N)$ represent the global outcomes and O is the overlap matrix of the POVM operators. However, direct reconstruction often leads to negative eigenvalues ($\rho \not\geq 0$) which results in a matrix that is mathematically incorrect as a quantum state and completely unphysical.

Note

We index all global outcomes by a single scalar $a = 1, \dots, 4^N$, writing n_a , p_a , and M_a instead of $n_{\vec{a}}$, $p_{\vec{a}}$, and $M_{\vec{a}}$.

In this context, Maximum Likelihood Estimation (MLE) searches for the statistically optimal — and physically valid — density matrix ρ that maximizes the probability of observing the collected experimental samples. To ensure the reconstruction of a valid quantum state (Hermitian $\rho = \rho^\dagger$ and positive semi-definite $\rho \geq 0$) MLE re-parameterizes the density matrix using the Cholesky decomposition [8]:

$$\rho(\vec{t}) = \frac{T(\vec{t}) T^\dagger(\vec{t})}{\text{Tr}[T(\vec{t}) T^\dagger(\vec{t})]} \quad (9)$$

where $T(\vec{t})$ is a lower-triangular matrix whose entries are parameterized by a real vector of parameters $\vec{t}$ (Equation 10). Here the product $TT^\dagger$ is positive semi-definite and Hermitian by construction for any choice of T , and dividing by the trace guarantees the strict normalization constraint ($\text{Tr}(\rho) = 1$).

$$T(\vec{t}) = \begin{pmatrix} t_1 & 0 & 0 & \dots & 0 \\ t_2 + it_3 & t_4 & 0 & \dots & 0 \\ t_5 + it_6 & t_7 + it_8 & t_9 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \dots & \dots & \dots & \dots & t_{d^2} \end{pmatrix} \quad (10)$$

Using this parametrization, the optimizer works directly on the unconstrained real parameters $\vec{t}$, ensuring the reconstruction of a physically valid state.

From a statistical point of view, if we collect $N_s = \sum_a n_a$ shots with n_a counts for each possible outcome a , assuming each experiment is i.i.d., the joint probability of $\{n_a\}$ follows a Multinomial distribution

$$P(\{n_a\} | \vec{t}) = \frac{N_s!}{\prod_{a=1}^{4^N} n_a!} \prod_{a=1}^{4^N} [p_a(\vec{t})]^{n_a} \quad (11)$$

where each $p_a(\vec{t})$ can be reconstructed using the Born Rule with the new parametrized $\rho(\vec{t})$. At this point, taking the logarithm and ignoring constant additive terms yields the Negative Loglikelihood (NLL) [8]:

$$\mathcal{L}(\vec{t}) = - \sum_{a=1}^{4^N} n_a \ln [\text{Tr}(M_a \rho(\vec{t}))] \quad (12)$$

The MLE estimate is then obtained by minimizing $\mathcal{L}(\vec{t})$ over the unconstrained parameter vector $\vec{t}$:

$$\vec{t}_{\text{opt}} = \arg \min_{\vec{t}} \mathcal{L}(\vec{t}) \quad (13)$$

The optimal density matrix is then recovered as $\rho(\vec{t}_{\text{opt}})$ via the Cholesky parametrization defined in Equation 10.

2.3. Generative Modeling via Variational Autoencoders

Variational Autoencoders are powerful generative models used in machine learning (ML) to generate new data in the form of variations of the input data on which they are trained [6].

Like all the autoencoders, VAEs are deep learning models composed of an *encoder* that learns to isolate the most important latent variables from the training data and a *decoder* that then uses those latent variables to reconstruct the input data. However, while most autoencoder architectures encode a discrete, fixed representation of latent variables, VAEs encode a continuous, probabilistic representation of that *latent space*, allowing not only to accurately reconstruct the exact original input but also to use variational inference to generate new data samples that resemble the original input data.

As anticipated, the VAE consists of two neural networks: an encoder q_ϕ and a decoder p_θ . Given a training sample $\mathbf{x}$, the encoder outputs a mean vector $\boldsymbol{\mu}$ and a standard deviation vector $\boldsymbol{\sigma}$, which parametrize a Gaussian distribution over the latent space. By using the reparametrization trick [6] a latent vector is sampled as $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$, allowing gradients to backpropagate through the sampling step, then the decoder maps $\mathbf{z}$ back to a reconstructed sample $\tilde{\mathbf{x}}$ as reported in Fig. 1b. Training proceeds by minimizing the negative Evidence Lower Bound (ELBO):

$$J(\theta, \phi; \mathbf{x}) = \underbrace{-\mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction Loss}} + \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))}_{\text{KL Divergence}}, \quad (14)$$

The ELBO is a tractable lower bound on the marginal log-likelihood $\log p_\theta(\mathbf{x})$, which is itself intractable due to the integral over the latent space; maximizing it therefore indirectly maximizes the data likelihood. The first term penalizes poor reconstruction, while the second regularizes the latent space by encouraging $q_\phi(\mathbf{z}|\mathbf{x})$ to stay close to the standard Gaussian prior $p(\mathbf{z}) = \mathcal{N}(0, I)$, preventing overfitting.

In our specific case the method consist of two steps: the training process and the generating process. The training procedure for the VAE is summarized in Fig. 1a. The training set is built from N_s simulated measurements of the target quantum state. At every training step, a batch of samples is drawn at random from this set, and the VAE parameters are updated via gradient descent on the loss function (Eq. 14); training continues until convergence.

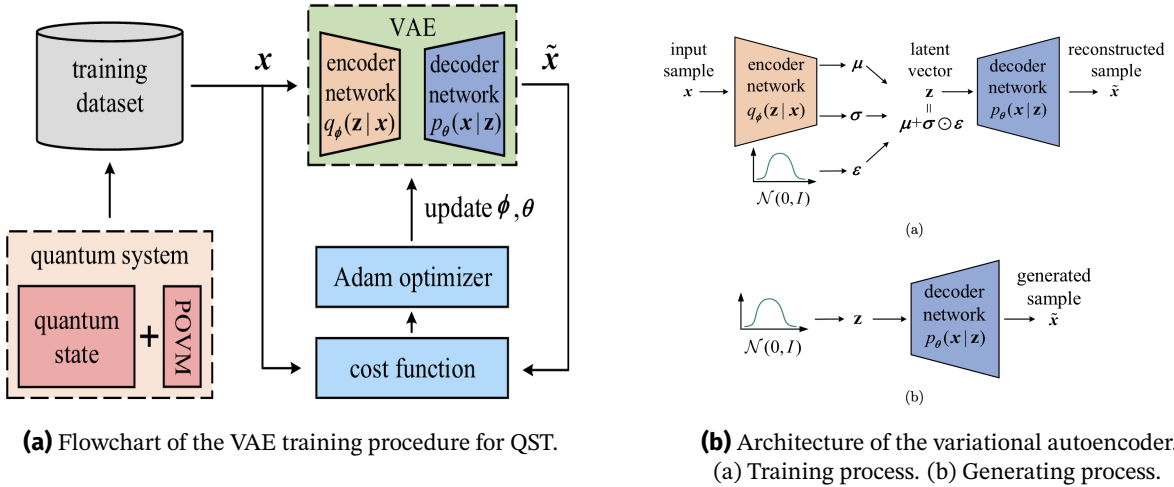


Figure 1. VAE structure and architecture used for QST. These images were taken from C.Chen et al. [5].

Once trained, the decoder is used as a generative model to estimate the VAE distribution P_{VAE} over the 4^N measurement outcomes. We draw N_{gen} latent vectors from the prior, decode them into per-qubit probability distributions, and sample one discrete outcome per qubit. The empirical frequencies are accumulated over the complete outcome support and normalized to yield P_{VAE} . The key point is that the generation stage is purely simulated and therefore N_{gen} can be chosen arbitrarily large. The Monte Carlo error on the estimated frequencies scales as $\propto 1/\sqrt{N_{\text{gen}}}$, so by taking $N_{\text{gen}} \gg N_{\text{train}}$ this sampling error is driven well below the shot noise of the training data and becomes negligible. As a consequence, any residual discrepancy in the reconstructed probability reflects the representational bias of the model rather than statistical fluctuations of the generation step, which makes the VAE-MLE comparison meaningful. From this by applying Equation 7 classical fidelity can be assessed.

3. NUMERICAL IMPLEMENTATION AND SIMULATION SETUP

Here, a brief but rigorous technical description of our Python-based setup is provided.

Data simulation. Quantum states were prepared by simulating the target circuits using Qiskit [7], which provides exact statevector simulation and the resulting true density matrix ρ . In this work, we consider three paradigmatic different states representing distinct entanglement regimes. For the case $N = 3$ the configurations are reported in Table 1.

State	Entanglement Level	State Vector
GHZ	Maximally entangled	$ \text{GHZ}_3\rangle = \frac{1}{\sqrt{2}}(000\rangle + 111\rangle)$
W	Distributedly entangled	$ W_3\rangle = \frac{1}{\sqrt{3}}(100\rangle + 010\rangle + 001\rangle)$
Product	Fully separable	$ +++\rangle = \frac{1}{2\sqrt{2}}(0\rangle + 1\rangle)^{\otimes 3}$

Table 1. State Configurations.

Once we have the true density matrix, the exact POVM outcome probability distribution $P_{\text{exact}} = \{p_a\}$ was computed analytically via the Born rule (Equation 5), and N_s discrete outcomes were sampled directly from P_{exact} , simulating² the statistical nature of real quantum measurements in an ideal, noiseless setting.

MLE. Regarding the MLE pipeline, we construct a custom negative log-likelihood function from the generated samples using the formulation in Equation 12 provided in Section 2. This function is subsequently minimized using the MIGRAD algorithm via the iminuit [4] Python package, yielding the optimal $\vec{t}_{\text{opt}}$, which is then used to compute the reconstructed density matrix $\rho(\vec{t}_{\text{opt}})$ under the Cholesky parametrization.

VAE. The VAE architecture was implemented as a single custom class subclassing `keras.Model`, which gives explicit control over the forward pass, the loss assembly and the training loop managed with `Tensorflow` [2, 3]. The encoder q_ϕ takes the $4N$ -dimensional input — corresponding to the one-hot encoded representation of the N qubit states — and passes it through two fully connected hidden layers with LeakyReLU³ activation functions, followed by two parallel linear heads that output the latent mean μ and the log-variance $\log \sigma^2$. The latent vector is then obtained through the reparametrization trick described above, and the decoder p_θ mirrors the encoder. Its final linear layer produces $4N$ logits, which are reshaped into N groups of four and passed through a per-group softmax, so that the reconstruction $\tilde{x}$ is, for each qubit, a normalized probability distribution over the four IC-POVM outcomes, consistent with the one-hot encoding of the training data. The two terms of the loss (Equation 14) are computed explicitly. The reconstruction term is evaluated as the Categorical Cross-Entropy between input and reconstruction, summed over the N qubits; the KL term is computed in closed form for the Gaussian posterior against the $\mathcal{N}(0, I)$ prior and the two averaged over the batch. A custom training step performs the forward pass under a gradient tape and updates the weights with the Adaptive Moment Estimation (Adam) optimizer. The layer widths h , latent dimension l and batch size are scaled with the number of qubits following the configuration of Chen et al. [3], as reported in Table 2.

N	Layout (i-h-h-l)	Batch Size	N_{gen}
3	12-96-96-16	100	20000
4	16-128-128-32	200	100000
6	24-1280-1280-256	600	$4^6 * 500$
8	32-2560-2560-512	1000	$4^7 * 500$

Table 2. Parameters configuration.

Moreover we applied callbacks such as `ReduceLROnPlateau` function as learning rate annealing starting from $lr = 0.001$ and `EarlyStopping` over the loss. In addition we used a `KLWarmup` custom callback to implement a *annealing schedule* (or warmup) for the β weight of the KL divergence term: it gradually linearly increases to its final target value ($\beta_{\text{max}} = 0.85$) over a set number of epochs. By dynamically increasing β , we allow the model to first focus entirely on learning how to encode and decode the basic features of the quantum state (maximizing reconstruction accuracy). Once a coherent representation is established, the gradual activation of the KL penalty smoothly regularizes the latent space into a continuous Gaussian structure without destroying the learned features.

²On standard quantum hardware, only projective measurements (PVMs) can be directly executed via native gates. Implementing a generalized POVM requires coupling the system to auxiliary ancilla qubits and performing higher-dimensional projective measurements. However, in this numerical simulation context, we bypass this hardware overhead by sampling directly from the exact analytical distribution P_{exact} .

³Leaky ReLU is a simple variant of classical ReLU that gives to negative values a little slope, preventing the "dying ReLU" problem.

Fidelity. To evaluate the accuracy of each reconstruction, we compute fidelities using the expressions introduced in Section 2. For MLE, the quantum fidelity F_Q provides a direct measure of overlap in Hilbert space; however, to enable a fair comparison with VAE — which never explicitly reconstructs ρ — we also apply the Born rule (Equation 5) to the MLE-reconstructed density matrix to obtain the corresponding *exact*⁴ POVM outcome distribution, and compute the same metric F_C on both methods. As additional baselines, we compute the classical fidelity of the raw empirical frequency distribution, which benchmarks the shot-denoising capabilities of each method, and — for the VAE — the classical fidelity of a uniform distribution over all outcomes, representing the fully agnostic prior at initialization.

All results are averaged over multiple independent runs with different datasets and random seeds. We report 95% percentile interval rather than standard deviations, since fidelity values are bounded above by 1 and the resulting distributions can be asymmetric near convergence.

The full implementation, including simulation pipelines and analysis notebooks, is available at [10].

4. RESULTS AND DISCUSSION

4.1. Performance and Scaling Limits of Classical MLE

Firstly, we characterize the MLE approach focusing on the maximally entangled GHZ state.

Considering a 3-qubit system — which guarantees tractable minimization runtime — we evaluate the convergence of the quantum and classical fidelities computed from the reconstructed density matrices as a function of the sample size N_{samples} . The corresponding results are reported in Fig. 2a. The correlation between these two variables highlights how, despite being less informative, classical fidelity provides an easily accessible and computationally cheap proxy to monitor the reconstruction quality. This confirms that minimizing the statistical contrast in the probability space effectively drives the estimator toward the true density matrix within the Hilbert space, thus ensuring that F_C serves as a suitable comparative metric within the proposed methodology.

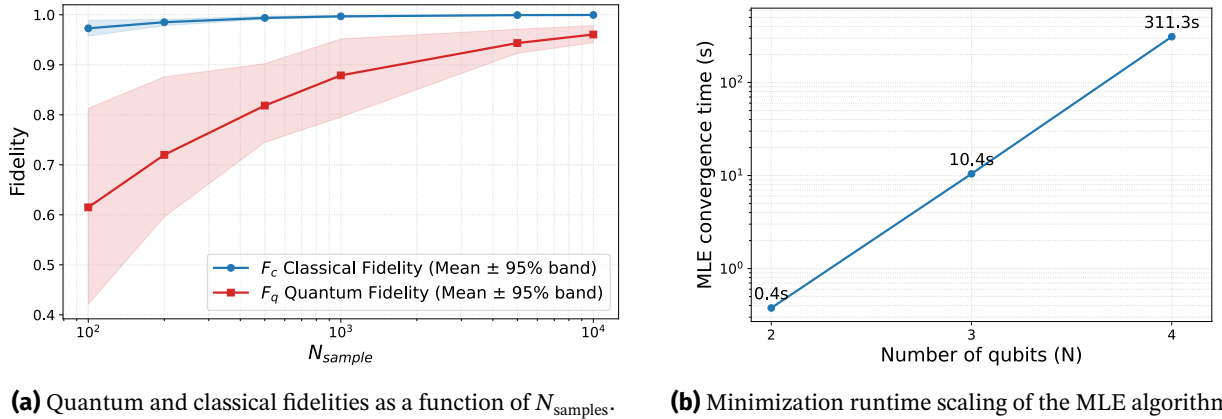


Figure 2. (a) Convergence behavior showing the reconstruction fidelity across different measurement sample sizes. Error bands were computed mediating over $B=50$ datasets and assessing 95% percentile. (b) Computational execution time exhibiting exponential growth as the system size scales; sample size is fixed at $N_{\text{sample}} = 500$.

Unfortunately, as previously anticipated, the MLE analysis is limited to small system sizes, such as the 3-qubit case. To highlight this limitation, Fig. 2b illustrates the execution time scaling as a function of the number of qubits. Already at 4 qubits, minimizing the negative log-likelihood function takes approximately 5 minutes per single run, rendering the evaluation of multiple statistical instances computationally prohibitive, especially if compared to VAE runtime. This numerical benchmark should be interpreted as a direct empirical demonstration of the curse of dimensionality. Although the minimization routine itself has not been optimized, the true bottleneck of the MLE method is due to the explicit reconstruction of the entire density matrix and the subsequent computation of the outcome probabilities within the NLL function. Indeed, this process demands storing and manipulating 4^N POVM operators — one for each possible multi-qubit measurement outcome — leading to an exponential memory footprint and computational overhead that quickly becomes intractable as N increases⁵. This scalability constraints motivates the hypothesis of the VAE framework as a solution, which bypasses the need of the full density matrix reconstruction during inference.

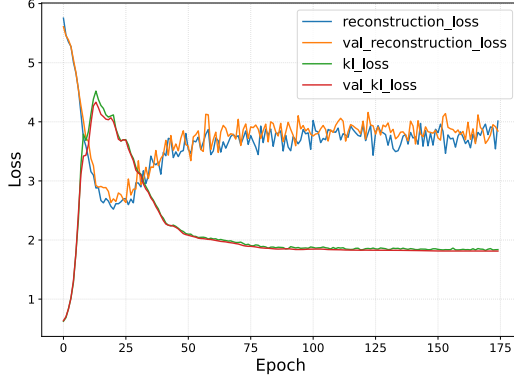
⁴In this case, each value is derived analytically from the reconstructed ρ ; no empirical sampling is involved, unlike the VAE approach.

⁵We face the exact same bottleneck during the synthetic data simulation phase of our framework. Because we generate the training samples by sampling directly from the exact, full-dimensional density matrix (ρ) using the Born Rule, this simulation scales exponentially, representing the primary constraint on our VAE’s scalability analysis to a higher number of qubits.

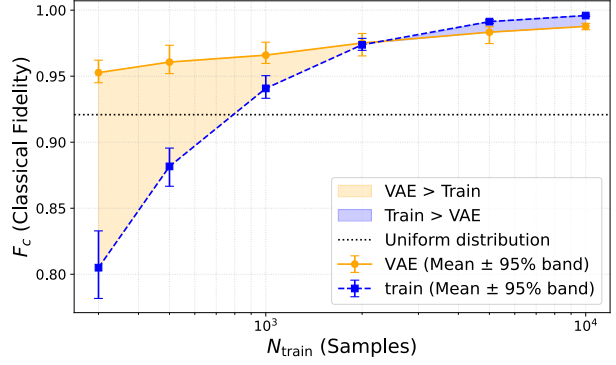
4.2. Quantum State Learning via VAE

We analyze the VAE in detail on the 4-qubit GHZ state, whose IC-POVM outcome space already contains $4^4 = 256$ distinct outcomes. This dimensionality is large enough to make the empirical distribution genuinely undersampled at moderate N_{train} and therefore an ideal test for assessing the denoising ability of the generative model.

In Fig. 3a are reported the training and validation curves for a representative run ($N_{\text{sample}} = 1000$). The behaviour confirms that the annealing schedule successfully avoids posterior collapse leading to non-pathological loss curves. During the warm-up phase, where the KL weight ramps linearly from 0 to $\beta_{\text{max}} = 0.85$, the reconstruction loss drops steeply, this indicates the model first exploits the latent code as a quasi-autoencoder. Crucially, the KL divergence does not decay to zero — it grows from ~ 0.4 and stabilizes around ~ 1.8 — which is the signature of an informative latent space. Once β saturates, the KL term acts as a regularizer that keeps the posterior close to the prior, while training/validation curves track each other closely, indicating no overfitting.



(a) VAE loss curves.



(b) Compare classical fidelity curves on training and reconstructed sample.

Figure 3. (a) Plot of reconstruction loss and KL divergence. (b) Classical fidelity plot over number of training samples. Error bands were computed mediating over $B=20$ datasets and assessing 95% percentile.

Then we executed multiple trainings starting from different datasets — generated from different seeds — by varying the number of training samples in order to evaluate how the training classical fidelity differs from the reconstructed sample one. Results are reported in Fig. 3b and the comparison identifies a clear advantage at low statistics: in the data-scarce regime the VAE substantially outperforms the training empirical distribution. The two curves cross around $N_{\text{sample}} = 2000$, beyond which the empirical estimate, slightly overtakes the model. It is interesting to note that the VAE fidelities stay always above the uniform-distribution floor fidelity situated at $F_c \approx 0.92$, indicating that the VAE model is able to learn the underlying physics of the system. The practically interesting window is therefore the low N_{sample} region, where the network replaces a number of expensive physical measurements with generative denoising. To inspect even more the learning process of the network, we study the outcome-level deviations as reported in Fig. 4.



Figure 4. Outcome-level deviations from p_{exact} at fixed $N_{\text{train}} = 1000$: the minor deviation between *empirical frequencies of sample training* and *reconstructed probability of VAE* compared to the exact probability are highlighted with markers and vertical lines. The outcomes are sorted by decreasing value of amplitude.

We can easily see the tendency of the VAE to underestimate high-probability outcomes and overestimate lower-probability ones. This behavior is highly coherent with a model that starts with no prior knowledge (acting effectively as a uniform generator) and gradually adapts to the distribution suggested by the training data. The VAE’s loss function forces the network to find a balance between memorizing the training samples (reconstruction loss) and aligning its latent space with an agnostic prior (KL divergence): as a result, when the model adapts to the probabilities suggested by the training data, the regularizer acts as a smoothing constraint. It pulls the highest peaks downwards and fills in the undersampled tails, preventing the network from overfitting to the empirical shot noise.

Finally, we evaluate the framework on GHZ states up to 8 qubits as reported in Fig. 5. The VAE performance remains stable as the Hilbert space grows, and the denoising advantage widens dramatically with system size, precisely because the empirical distribution becomes dramatically undersampled over an exponentially larger outcome space.

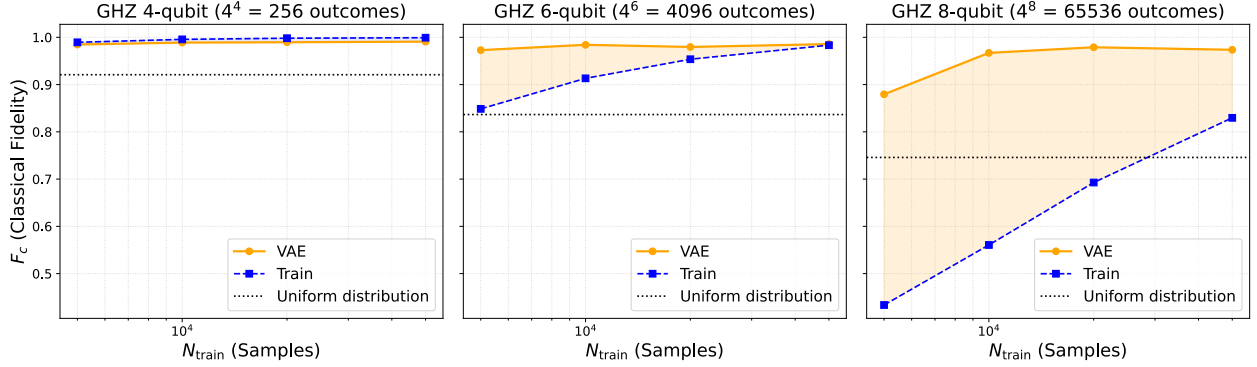


Figure 5. Classical fidelity of different number of qubits over different sample size. Results are related to single runs due to limitations in computational runtime.

With $N = 8$ and $N_{\text{samples}} = 10,000$, the raw empirical fidelity collapses to $F_C = 0.56$, whereas the VAE still reconstructs the state at $F_C = 0.97$. It is clear the network generalizes effectively in regimes where the direct empirical estimate is meaningless. Critically, we observe that the primary limiting factor of this framework at high qubit numbers (N) is the evaluation of the generated distribution P_{VAE} itself, which is required to compute the classical fidelity. To keep the Monte-Carlo noise on P_{VAE} negligible over an outcome space of size 4^N , the number of generated samples scales significantly with the system size, as reported in Table 2⁶). The steep growth of N_{gen} , combined with the wider networks used at larger qubits number, makes this generation step dominate the runtime. In other words, the cost of scaling this approach is the price of its accuracy: resolving a distribution over an exponentially large support with low statistical noise requires an exponentially large number of decoder samples.

Note

It is crucial to distinguish this model-intrinsic limitation from an external bottleneck encountered in our study, which is strictly related to the data simulation phase rather than the generative framework itself. Producing the ground-truth reference and sampling the IC-POVM outcomes via Born rule requires handling a probability table that scales as 4^N . This means the limitations of our scaling analysis is also due to the simulation of the measurement statistics, not only to the generative model itself.

4.3. Comparative Methodological Analysis: MLE vs. VAE

We perform a direct comparison of the two reconstruction methods on the 3-qubit systems reported in Table 1. To obtain a robust statistical comparison we evaluate $B = 50$ independent datasets per sample size, with $N_{\text{sample}} \in \{100, 200, 500, 1000, 5000, 10000\}$, and summarize each configuration by its mean F_C and the 95% percentile band of each distribution.

In Fig. 6 we presents the comparison of F_C versus N_{sample} , overlaying the MLE and VAE curves with a shared F_C axis and the ΔF_C of VAE with respect of MLE. The shared axis shows that the relative ordering of the two methods is not universal but depends on the physical state. The grid separates three behaviours: MLE strictly above VAE for GHZ, a markedly wider separation for W, and two overlapped curves for the product state.

The most informative outcome of the comparison is how the entanglement structure affects each method differently.

⁶Fairly speaking, for 8-qubit case we didn’t use $4^8 * 500$, instead $4^7 * 500$ due to limited computation resources.

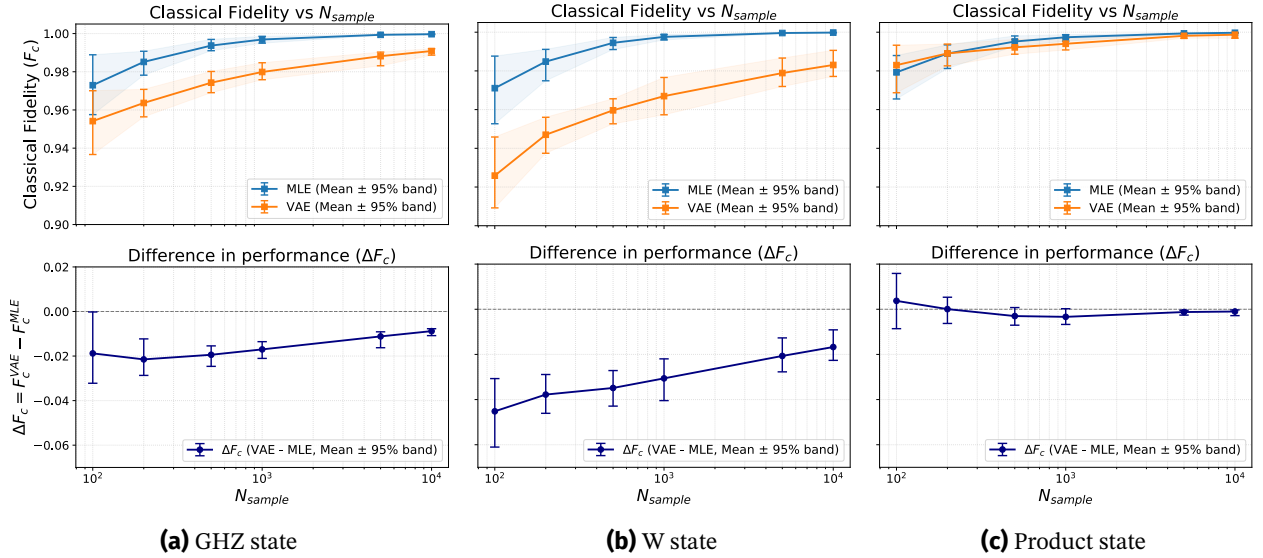


Figure 6. Upper plots report both fidelity approach for GHZ, W and product state respectively. Lower plots report ΔF_c between VAE and MLE. Error bands were computed mediating over $B=50$ datasets and assessing 95% percentile.

MLE is essentially insensitive to the state. It reconstructs the full density matrix by maximum likelihood and converges to unit fidelity for all three states, with no measurable dependence on whether the state is maximally, partially, or non-entangled.

The VAE, in contrast, is strongly modulated by the state’s complexity. Its accuracy is highest on the product state — where it matches MLE and even slightly exceeds it at the lowest statistics — degrades on GHZ, and is worst on W. This ordering follows directly from the decoder’s architecture: the output is factorized into independent per-qubit softmaxes, so a factorizable (product) distribution is represented exactly with negligible demand on the latent sapce, whereas every inter-qubit correlation must be encoded in the latent variables. The richer and more distributed the correlation structure — minimal for $|++\rangle$, concentrated for GHZ, spread across many outcomes for W — the harder the encoding task and meaning of latent variables, and the larger the bias in reconstructed distribution.

5. CONCLUSION

In conclusion we highlight that the comparison delineates a clear division of roles. MLE is the method of choice at low qubit count: it is statistically exact, delivers the highest fidelity across all states and all sample sizes, is insensitive to entanglement structure, and reconstructs the complete density matrix rather than only the outcome distribution, giving access also to quantum fidelity. Its cost, however, grows exponentially: the number of free parameters scales as $4^N - 1$ and the optimization time explodes accordingly, making it intractable beyond a bunch of qubits. The VAE is the method of choice in the high dimensionality regime: it does not reconstruct the full state but provides tomography just to a distribution level, it remains accurate and stable up to $N = 8$ — where MLE is intractable — and its advantage over the raw empirical distribution grows with the system size. The price is a lower fidelity at small N , a bias toward factorizable correlation structures, and its own generation bottleneck. We conclude that MLE is better for statistical exactness and full reconstruction at low qubit count while the VAE is preferred for computational scalability and fidelity tomography in high dimension.

In this work were considered only pure states as states to be reconstructed. However, as already mentioned, real systems are characterized by noisy hardwares due to the interaction with the external environment. For future works it would be interesting to extend this systematic comparison to mixed state —subjected to general noise models— to assess how physical noise would affect these results.

Appendix A: BLOCH SPHERE AND SIC-POVM

The Bloch sphere provides a geometrical representation of a single two-level quantum system (qubit). Any associated density matrix ρ can be expressed in terms of the Pauli vector $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ as:

$$\rho = \frac{\mathbb{I} + \vec{r} \cdot \vec{\sigma}}{2}$$

where $\vec{r} = (x, y, z) \in \mathbb{R}^3$ is the Bloch vector (see Fig. 7a) and its norm $|\vec{r}|$ directly measures the state purity ($|\vec{r}| = 1$ for pure states, $|\vec{r}| < 1$ for mixed states).

In this work, we directly simulate a Symmetric Informationally Complete POVM (SIC-POVM) [5]. This tetrahedral configuration consists of four elements $\{M_\alpha\}_{\alpha=1}^4$ pointing toward the vertices of a regular tetrahedron inscribed in the Bloch sphere (see Fig. 7c):

$$M_\alpha = \frac{1}{4} (\mathbb{I} + \vec{s}_\alpha \cdot \vec{\sigma})$$

where the explicit vertex vectors $\vec{s}_\alpha$ are:

$$\vec{s}_1 = (0, 0, 1), \quad \vec{s}_2 = \left(\frac{2\sqrt{2}}{3}, 0, -\frac{1}{3}\right), \quad \vec{s}_3 = \left(-\frac{\sqrt{2}}{3}, \sqrt{\frac{2}{3}}, -\frac{1}{3}\right), \quad \vec{s}_4 = \left(-\frac{\sqrt{2}}{3}, -\sqrt{\frac{2}{3}}, -\frac{1}{3}\right)$$

Because these four operators span the 4-dimensional space of 2×2 Hermitian matrices, the statistics collected from this single configuration are mathematically sufficient to uniquely reconstruct any arbitrary qubit state.

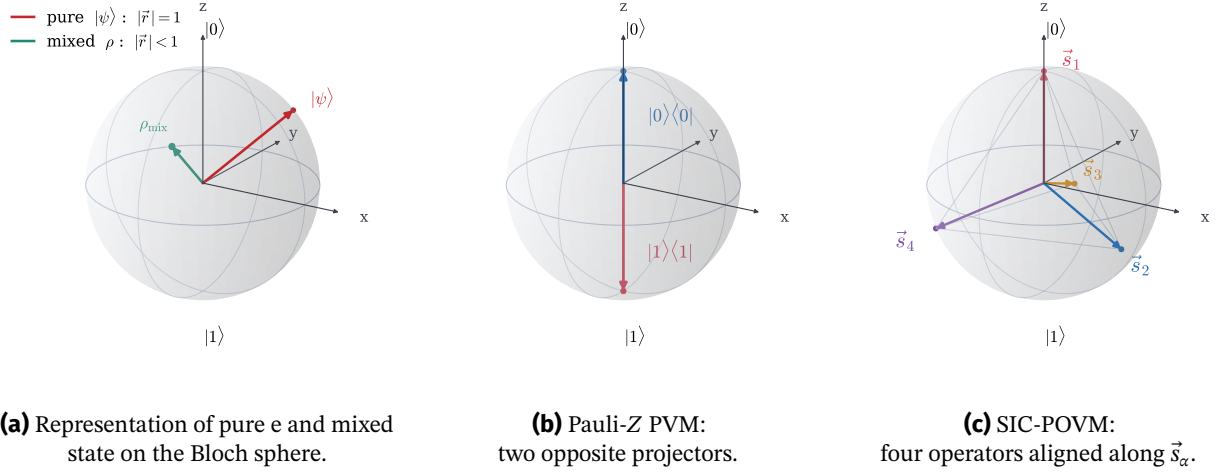


Figure 7. Bloch sphere representations of quantum states and measurement frameworks.

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